Towards a refined 3D slab model of Western Pacific to investigate the nature of deep earthquakes

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Introduction

The nature of deep earthquakes with depths greater than 70 km is enigmatic because brittle failure at this high-temperature and high-pressure regime should be inhibited. There have been three mechanisms proposed to explain what causes deep earthquakes in subduction zones, dehydration embrittlement, phase transformational faulting, and thermal runaway instability. The existing seismological constraints can't yet definitively distinguish between these three major hypotheses. One of the main reasons is that the fine 3D structures of subducting slabs are not well constrained in terms of slab upper interface, slab thickness, and slab internal fine layering. This study utilizes a large data set of seismic waves traveling through the Western Pacific subduction zones including Kuril, Japan, and Izu-Bonin. Our data set includes the seismic waveforms recorded by about 2,000 broadband stations in China, Japan, Korea, Mongolia, and other neighboring regions. These waveforms provide rich information of the 3D slab structures which can be retrieved by full waveform inversion. In order to get refined 3D slab models, we start with an initial 3D model that combines two previous full waveform inversion models of East Asia (FWEA18 and EARA2014), embedded in CRUST1.0 for the crustal model and S362ANI for the mantle model. We expect the final model, which will predict the waveforms to the shortest period of 9 seconds of the new data set, will help more definitively constrain the deep slab interface in the Western Pacific subduction zones therefore the relative spatial position between deep earthquakes and slab 3D structural details.

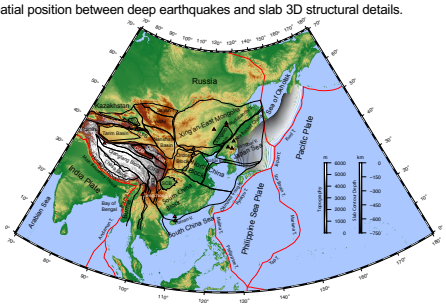

Figure 1. Tectonic and topographic map of East Asia. Red triangles mark active volcanoes. White, grey to dark semi-transparent lines indicate Japan and Kuril slab surface iso-depth from shallow to deep. Dark lines define the main tectonic units and basins. Abbreviations are as follows: CDB, Chuandian Block; JGB, Junggar Basin; QDB, Qaidam Basin; SCB, Sichuan Basin; TLFB, Tulufan Basin; SGFB, Songpan Ganzi Fold Belt. Red bold lines delineate plate boundaries from NUVEL-1 [DeMets et al., 1990].

Data and Methods

Full waveform inversion (FWI), i.e., adjoint tomography [Tromp et al., 2005] based on a highly accurate spectral element method [Komatitsch and Tromp, 2002a, 2002b], is applied to a large seismic waveform data set mainly from CEArray (Chinese Earthquake Administration Array) in China, the seismic network in Korea, and F-Net in Japan. The initial 3D model used for the inversion is a smoothed composite model combining FWEA18 [Tao et al., 2018) and EARA2014 [Chen et al., 2015] embedded in a global model with S362ANI [Kustowski et al., 2008] in the mantle and CRUST1.0 [Laske et al., 2013] in the crust.

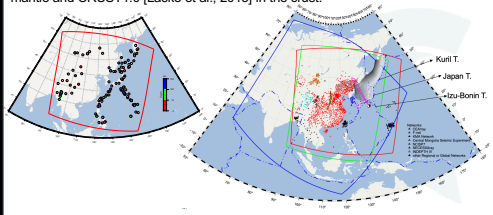

Figure 2. Event and stations distribution map. **Left** : A total of 130 events used in our inversion are marked by colored beach balls, selected with 5<Mw<6.7 between year 2009 – 2018. Beach ball color represents event depth. **Right**: A total of 2,013 stations are used in our inversion. The colored triangles mark stations from different networks. The red box encloses our region of interest, the green box the simulation region of model FWEA18, and the blue box the simulation region used by model EARA2014. The beach ball represents an example earthquake (global CMT event ID 201201010527A) in the Izu region with Mw of 6.8 and depth of 352 km. The waveform fitting between data and synthetics of this earthquake are shown in Figure 4 for the stations enclosed by the magenta lines.

Validation of the FWI initial model

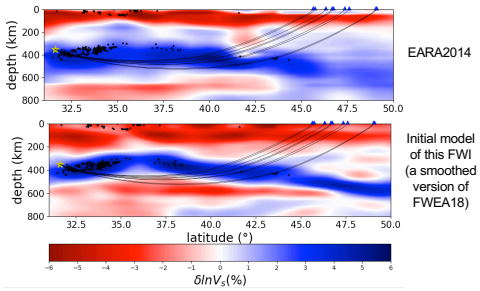

Figure 3. Comparison of Vs anomalies between the initial model (mostly smoothed FWEA18) and model EARA2014 for a cross section along the corridor marked by the magenta line in Figure 2. The yellow star denotes an example deep earthquake, the waveforms of which recorded by the stations (blue triangles) are shown in Figure 4. The black dots represent background seismicity from EHB catalog within 25-km distance of the cross section. Black lines are the theoretical ray paths predicted by model AK135.

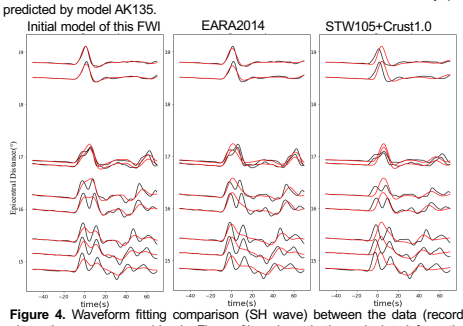

Figure 4. Waveform fitting comparison (SH wave) between the data (recorded along the magenta corridor in Figure 3) and synthetics calculated from three different models, initial model of this FWI (mostly smoothed FWEA18) (**left**), EARA2014 (**middle**), and STW105+Crust1.0 (**right**). Note that STW105 is a 1-D model. The slab structure in the initial model best predicts the diffracted waveforms.

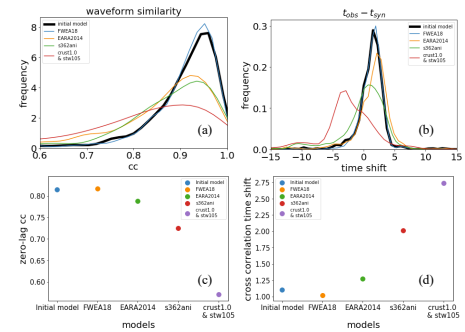

Figure 5. Comparison of statistical results of waveform fitting between data of 30 representative events in our study region and synthetics of five different models. **(a)** and **(b)** are the histograms of zero-lag cross correlation (ZLCC) (indicating waveform similarity) and travel time misfit. The measurement windows are manually selected based on the data waveforms. **(c)** and **(d)** are the model comparison of geographically weighted ZLCC and travel time misfit. The comparison shows that the initial model (a smoothed version of FWEA18) has similarly best predictability of the waveforms as FWEA18 in terms of both waveform similarity and travel time.

Earthquake Source Parameter Inversion

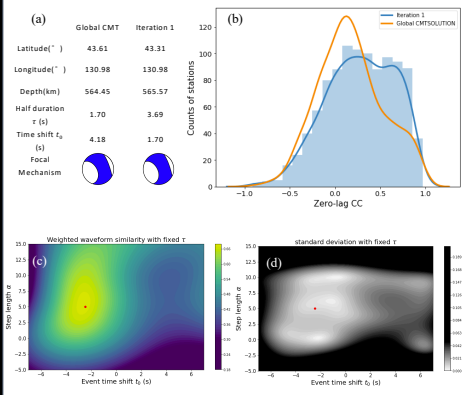

Figure 6. An example for the earthquake source parameter inversion (GCMT event ID: 201105101526A). The points source Fréchet kernels are calculated based on the adjoint method [Younghee Kim et al., 2011]. The waveform-similarity (zero-leg cross-correlation) based misfit is used for the gradient calculation of source parameters (i.e., location, moment tensor, origin time and source duration) [Tao et al, 2018]. Source parameters are updated iteratively by searching along the negative misfit gradient direction. Optimal step length (α), event origin time shift (t_0), and source duration (τ) are searched at each iteration with Bayesian optimization method in this study. **(a)** The table lists the source parameters before and after the source inversion. **(b)** The histograms show the waveform similarity improvement after the first iteration. The waveform similarity distribution is displayed as function of α and t_0 based on Bayesian optimization **(c)** The figure shows the weighted similarity distribution. **(d)** The corresponding standard deviation of (c). The red dot in (c) and (d) marks where the optimal parameters are located.

Sensitivity kernels of different body waves

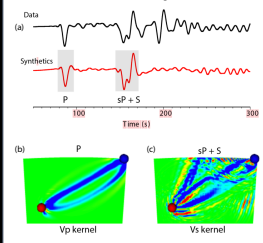

Figure 7. Data and 3D synthetics comparison for the vertical component record of station HRO and the corresponding P- and S-wave banana-doughnut kernels. **(a)** Data (black) and synthetics (red) are bandpass filtered between 6–150 s, and vertical cross-sections of **(b)** P- and **(c)** S-wave banana-doughnut kernels for station HRO (Δ = 5.74°) and event 200609160222A (depth = 318.6 km), are calculated by using the adjoint sources constructed from the highlighted grey windows in (a).

Future work

The initial 3D model and the updated structural models after every five iterations will be used to update and improve the source parameters of the earthquakes. The data-synthetic waveform misfits based on the zero-lag cross correlation are incorporated to calculate adjoint sources in the period band of 10 –120 seconds for body waves and 40 –120 seconds for surface waves. The Fréchet kernels of two shear wave speeds, Vsh and Vsv, and bulk sound speed, Vc, will be constructed to iteratively refine the 3D slab structural models. The new model will provide more robust seismological constraints of the Pacific slab upper interface geometry and the thickness of the slab. We expect the new seismic wave speed constraints and more accurate deep earthquake hypocenter locations will help us better understand the mechanism of the deep intraslab earthquake in high-temperature and high-pressure regime.

Acknowledgements

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